Hidden Markov Model (HMM)

Hidden Markov Model needs

- Initial probability
- Transition probability
- Observation probability

Example:

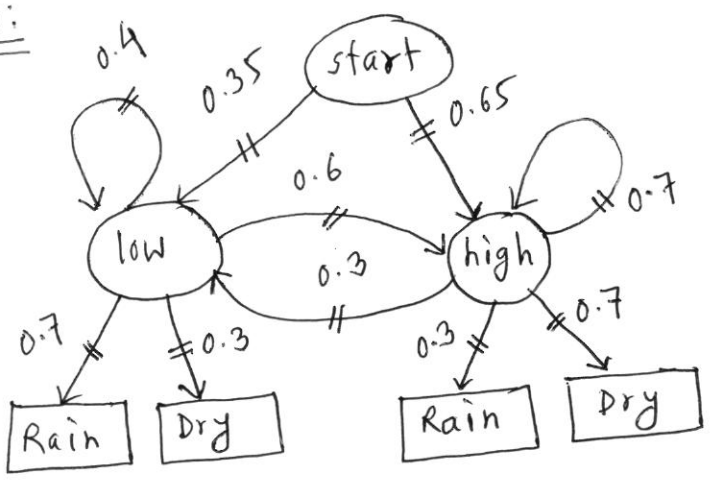
For forecasting weather, it is observed that low atmospheric pressure causes a chance of 70% Rain, otherwise Dry. On the other hand, 70% chance of Dryness is happened if high atmospheric pressure prevails. Again, there is a 60% probability of changing from low to high atmospheric pressure, but the high to low changing occurs in a probability of 30%. The initial probability of high atmospheric pressure is 65%.

Answer the following questions

a) Draw a hidden Markov model (HMM)

b) If rainy weather prevails on today, take a decision whether the cricket match on ~~the day immediate after~~ tomorrow will be happened or not.

a) Ans:



Initial probability:

$$P(\text{high} \mid \text{start}) = 0.65$$
$$P(\text{low} \mid \text{start}) = 0.35$$

Transition probability:

	low	high
low	0.4	0.6
high	0.3	0.7

$$P(\text{low} \mid \text{low}) = 0.4, \quad P(\text{high} \mid \text{low}) = 0.6$$
$$P(\text{low} \mid \text{high}) = 0.3, \quad P(\text{high} \mid \text{high}) = 0.7$$

observation probability:

$$P(\text{Rain} \mid \text{low}) = 0.7 \quad | \quad P(\text{Rain} \mid \text{high}) = 0.3$$
$$P(\text{Dry} \mid \text{low}) = 0.3 \quad | \quad P(\text{Dry} \mid \text{high}) = 0.7$$

b) Ans:

$$P(\text{Rain Rain}) = ?$$
$$P(\text{Rain Dry}) = ?$$

For P(Rain Rain)

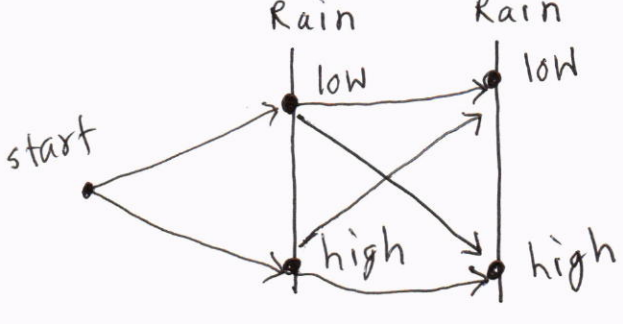


Figure 1

For P(Rain Dry)

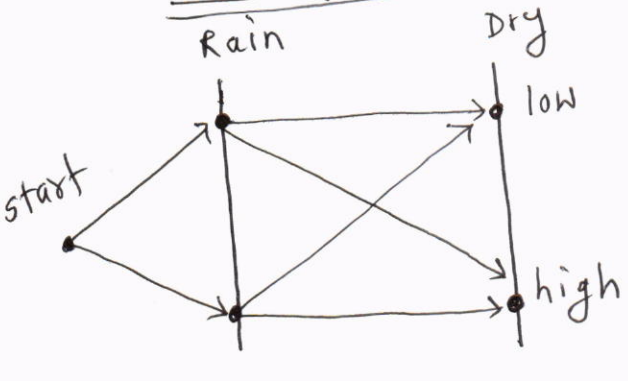


Figure 2

For Figure 1

Hidden state Sequence	Path Probabilities
start → low → low	P(low|start) * P(Rain|low) * P(low|low) * P(Rain|low) = 0.35*0.7*0.4*0.7
start → low → high	P(low|start) * P(Rain|low) * P(high|low) * P(Rain|high) = 0.35*0.7*0.6*0.3
start → high → low	P(high|start) * P(Rain|high) * P(low|high) * P(Rain|low) = 0.65*0.3*0.3*0.7
start → high → high	P(high|start) * P(Rain|high) * P(high|high) * P(Rain|high) = 0.65*0.3*0.7*0.3

Total probability = 0.1946

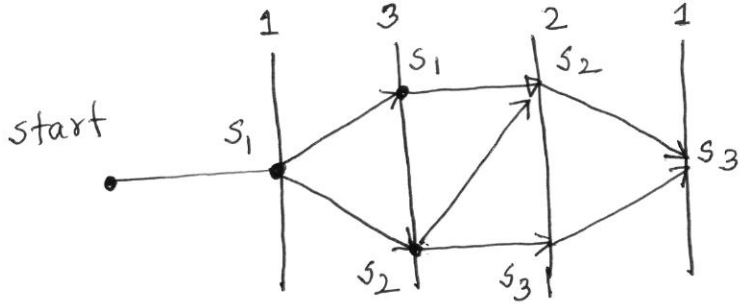
For Figure 2

Hidden state Sequence	Path Probabilities
start → low → low	P(low|start) * P(Rain|low) * P(low|low) * P(Dry|low) = 0.35*0.7*0.4*0.3
start → low → high	P(low|start) * P(Rain|low) * P(high|low) * P(Dry|high) = 0.35*0.7*0.6*0.7
start → high → low	P(high|start) * P(Rain|high) * P(low|high) * P(Dry|low) = 0.65*0.3*0.3*0.3
start → high → high	P(high|start) * P(Rain|high) * P(high|high) * P(Dry|high) = 0.65*0.3*0.7*0.7

Total probability = 0.2454

Since P(Rain Dry) > P(Rain Rain), so the cricket match will be held.

Ans:



HMM for character 'A':

Hidden state sequence	Path Probabilities
$s_1 \to s_1 \to s_2 \to s_3$	$P(s_1\|start) * P(1\|s_1) * P(s_1\|s_1) * P(3\|s_1) * P(s_2\|s_1) * P(2\|s_2) * P(s_3\|s_2) * P(1\|s_3)$ $= 1 * 0.9 * 0.8 * 0 * 0.2 * 0.8 * 0.2 * 0.9 = 0$
$s_1 \to s_2 \to s_2 \to s_3$	$P(s_1\|start) * P(1\|s_1) * P(s_2\|s_1) * P(3\|s_2) * P(s_2\|s_2) * P(2\|s_2) * P(s_3\|s_2) * P(1\|s_3)$ $= 1 * 0.9 * 0.2 * 0.8 * 0.8 * 0.2 * 0.9 = 0.00208$
$s_1 \to s_2 \to s_3 \to s_3$	$P(s_1\|start) * P(1\|s_1) * P(s_2\|s_1) * P(3\|s_2) * P(s_3\|s_2) * P(2\|s_3) * P(s_3\|s_3) * P(1\|s_3)$ $= 1 * 0.9 * 0.2 * 0.1 * 0.2 * 0.1 * 1 * 0.9 = 0.000324$

Total Probability = 0.0023976

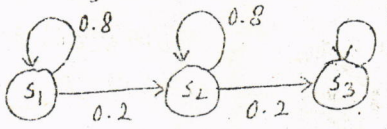
HMM for character 'B':

Hidden state sequence	Path Probabilities
$s_1 \to s_1 \to s_2 \to s_3$	0
$s_1 \to s_2 \to s_2 \to s_3$	0.0027648
$s_1 \to s_2 \to s_3 \to s_3$	0.006912

Total probability = 0.0096768

Therefore, character 'B' is more likely for given observation sequence {1, 3, 2, 1}

The structure of hidden states:



Transition probabilities: $\{a_{ij}\} =$

	S_1	S_2	S_3
S_1	0.8	0.2	0
S_2	0	0.8	0.2
S_3	0	0	1

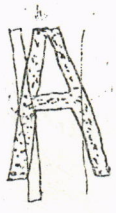
Observation:

observation = number of islands in the vertical slice

For character A:

observation probabilities: $\{b_{jk}\} =$

state \ observation			
S_1	(0.9)	(0.1)	0
S_2	0.1	0.8	0.1
S_3	0.9	0.1	0



For character B:

Observation probabilities:

state \ observation			
	0.9	0.1	0
	0	0.2	0.8
	0.6	0.4	0



Suppose that after character image segmentation the following sequence of island numbers in 4 slices was observed.

{1, 3, 2, 1}

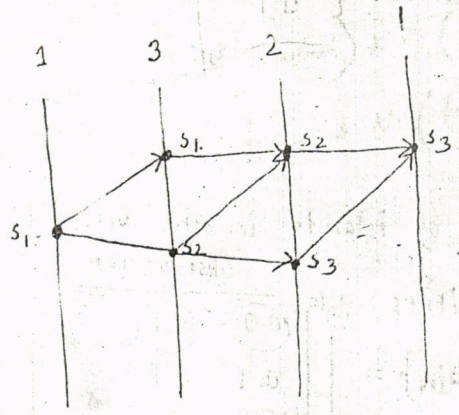
What HMM is more likely to generate this observation sequence, HMM for 'A' or HMM for 'B'?

Ans:

HMM for character 'A':

Hidden state Sequence	Transition Probabilities	Observation Prob	Total
$S_1 \to S_1 \to S_2 \to S_3$	$0.8 * 0.2 * 0.2 *$	$0.9 * 0 * 0.8 * 0.9$	0
$S_1 \to S_2 \to S_2 \to S_3$	$0.2 * 0.8 * 0.2 *$	$0.9 * 0.1 * 0.8 * 0.9$	0.00207
$S_1 \to S_2 \to S_3 \to S_3$	$0.2 * 0.2 * 1 *$	$0.9 * 0.1 * 0.1 * 0.9$	0.000324

Total = 0.0023976



For A

b_{jk} — j: state, k: observation

For calculating state seq 1

$P(S_1) * P(1|S_1) * P(S_1|S_1) * P(3|S_1) * P(S_1|S_1) * P(2|S_2) * P(S_3|S_2) * P(1|S_3)$

$= 1 * b_{11} * P(S_1|S_1) * b_{13} * P(S_2|S_1) * b_{22} * P(S_3|S_2) * b_{31}$

$= P(S_1|S_1) * P(S_2|S_1) * P(S_3|S_2) * b_{11} * b_{13} * b_{22} * b_{31}$

$= 0.8 * 0.2 * 0.2 * 0.9 * 0 * 0.8 * 0.9$

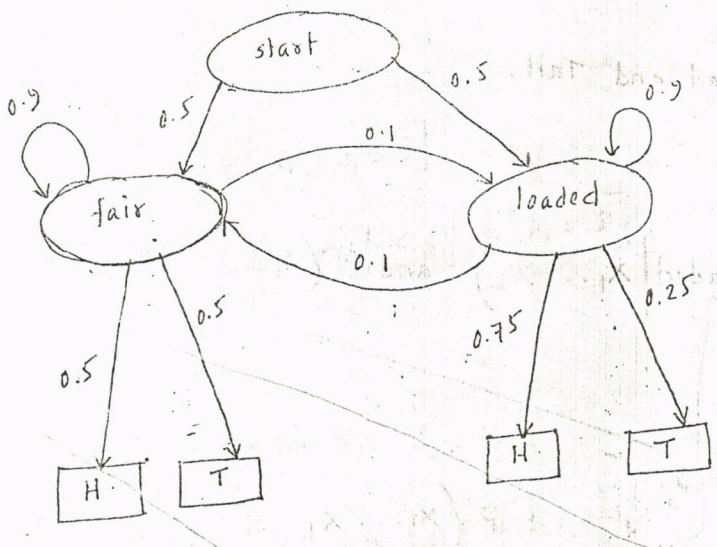
IMM for character 'B'

hidden state Sequence	Transition Probabilities	Observation Prob	Total
$S_1 \to S_1 \to S_2 \to S_3$	$0.8 * 0.2 * 0.2$	$* \; 0.9 * 0 * 0.2 * 0.6$	0
$S_1 \to S_2 \to S_2 \to S_3$	$0.2 * 0.8 * 0.2$	$* \; 0.9 * 0.8 * 0.2 * 0.6$	0.0027648
$S_1 \to S_2 \to S_3 \to S_3$	$0.2 * 0.2 * 1$	$* \; 0.9 * 0.8 * 0.4 * 0.6$	0.006912

Total = 0.0096768

Therefore, character 'B' is more likely for given observation sequence {1, 3, 2, 1}

Coin-Tossing Example: ***



⇐ Hidden states

⇐ observations

Transition Probabilities:

	fair	loaded
fair	0.9	0.1
loaded	0.1	0.9

Initial Probabilities:

$P(\text{fair}) = 0.5$

$P(\text{loaded}) = 0.9$



Emission or observation Probabilities:

- $P(H \mid \text{fair}) = 0.5$
- $P(T \mid \text{fair}) = 0.5$
- $P(H \mid \text{loaded}) = 0.75$
- $P(T \mid \text{loaded}) = 0.25$

that are the probabilities for Fair/Loaded coins given the set of tosses

$\{x_1, \ldots, x_L\}$ — sequence of Head and Tail.

Ans:

$P(s_i = \text{loaded} \mid x_1 \cdots x_L)$ and $P(s_i = \text{Fair} \mid x_1 \cdots x_L)$

s

$$P(s_i = s \mid x_1 \cdots x_L) = \frac{P(x_1 \cdots x_L, s_i = s)}{P(x_1 \cdots x_L)}$$

$= ???$

Where $P(x_1 \cdots x_L) = P(x_1 \cdots x_L, s_i = s_1) + P(x_1 \cdots x_L, s_i = s_2)$

$+ \cdots\cdots + P(x_1 \cdots x_L, s_i = s_L)$

$$= \sum_{s'} P(x_1 \cdots x_L, s_i = s')$$

]

[statistics Formula:

$$P(x_1 \cdots x_L, s_i = s) = P(x_1 \cdots x_i, s_i = s) * P(x_{i+1} \cdots x_L \mid x_1 \cdots x_i, s_i = s)$$

$$= P(x_1 \cdots x_i, s_i = s) * P(x_{i+1} \cdots x_L \mid s_i = s)$$

$f_i(s)$ — Foward algorithm

$b_i(s)$ — backward algorithm

Forward algorithm:

Compute $f_i(s) = P[x_1 \cdots x_i, s_i = s]$ for $i = 1 \ldots L$

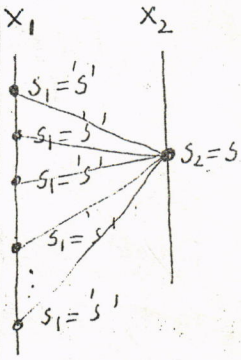
Base step:

$i=1$, $f_1(s) = P[x_1, s_i = s]$

$= P[s_i = s] * P[x_1 | s_1 = s]$

2nd step:

$i=2$, $f_2(s) = P[x_1 x_2, s_2 = s]$



$= P[x_1, s_1 = 's'] * P[s_2 = s | s_1 = 's'] * P[x_2 | s_2 = s]$

$+ P[x_1, s_1 = 's'] * P[s_2 = s | s_1 = 's'] * P[x_2 | s_2 = s]$

$+ \ldots\ldots$

$= \sum_{s'} \left(P[x_1, s_1 = 's'] * P[s_2 = s | s_1 = 's'] * P[x_2 | s_2 = s] \right)$

$= \sum_{s'} \left(f_1(s') * P[s | 's'] * P[x_2 | s] \right)$

forward probability · Transition probability · observation or emission probability

ith step:

$f_i(s) = \sum_{s'} \left(f_{i-1}(s') \; P(s|s') * P(x_i | s) \right)$

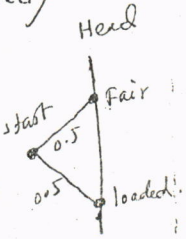
Find $P[S_i = \text{loaded} \mid \text{Head, Head, Tail})$ and $P[S_i = \text{fair} \mid \text{Head, Head, Tail})$ using forward algorithm.

Ans:

Forward algorithm First step:

$$f_1(\text{Loaded}) = P(\text{Head}, S_1 = \text{Loaded}) = P(\text{Loaded}) * P(\text{Head} \mid \text{Loaded})$$
$$= 0.5 * 0.75$$
$$= 0.375$$

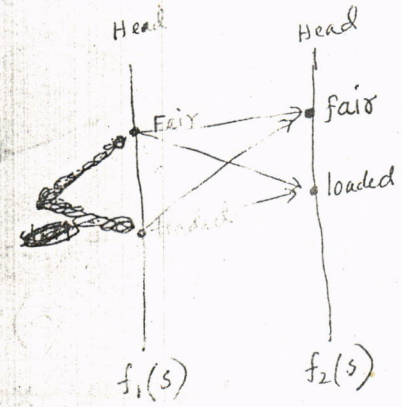
$$f_1(\text{fair}) = P(\text{Head}, S_1 = \text{Fair}) = P(\text{Fair}) * P(\text{Head} \mid \text{Fair})$$
$$= 0.5 * 0.5$$
$$= 0.25$$



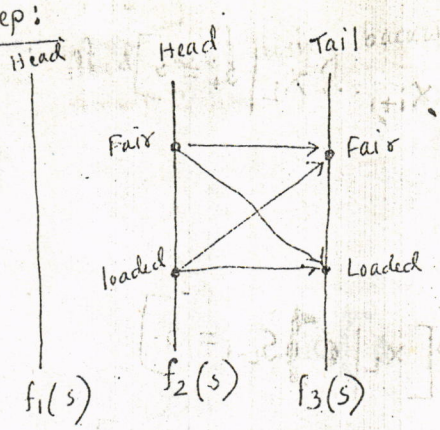
Forward algorithm second step:

$$f_2(\text{Loaded}) = P[X_1 X_2 = \text{Head Head}, S_2 = \text{Loaded}]$$
$$= f_1(\text{Loaded}) * P(\text{loaded} \mid \text{loaded}) * P(\text{Head} \mid \text{loaded}) + f_1(\text{fair}) * P(\text{loaded} \mid \text{fair}) * P(\text{Head} \mid \text{fair})$$
$$= 0.375 * 0.9 * 0.75 + 0.25 * 0.1 * 0.5 = 0.271875$$

$$f_2(\text{fair}) = P[X_1 X_2 = \text{Head Head}, S_2 = \text{fair}]$$
$$= f_1(\text{Loaded}) * P(\text{fair} \mid \text{loaded}) * P(\text{Head} \mid \text{fair}) + f_1(\text{Fair}) * P(\text{Fair} \mid \text{Fair}) * P(\text{Head} \mid \text{Fair})$$
$$= 0.375 * 0.1 * 0.5 + 0.25 * 0.9 * 0.5$$
$$= 0.13125$$



orward algorithm third step:



$$f_3(\text{loaded}) = P[x_1 x_2 x_3 = \text{Head Head Tail},\ s_3 = \text{loaded}]$$

$$= f_2(\text{loaded}) * P[\text{loaded} \mid \text{loaded}] * P[\text{Tail} \mid \text{loaded}]$$
$$+ f_2(\text{fair}) * P[\text{loaded} \mid \text{Fair}] * P[\text{Tail} \mid \text{loaded}]$$

$$= 0.271875 * 0.9 * 0.25 + 0.13125 * 0.1 * 0.25$$

$$= 0.06445$$

$$f_3(\text{Fair}) = P[x_1 x_2 x_3 = \text{Head Head Tail},\ s_3 = \text{Fair}]$$

$$= f_2(\text{loaded}) * P[\text{Fair} \mid \text{loaded}] * P[\text{Tail} \mid \text{Fair}]$$
$$+ f_2(\text{Fair}) * P[\text{Fair} \mid \text{Fair}] * P[\text{Tail} \mid \text{Fair}]$$

$$= 0.271875 * 0.1 * 0.5 + 0.13125 * 0.9 * 0.5$$

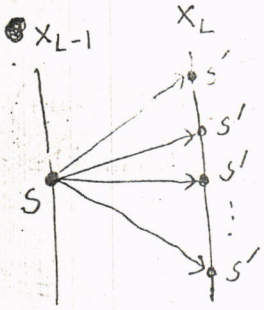
Backward algorithm

Compute $b_i(s) = P[x_{i+1} \cdots x_L \mid s_i = s]$ for $i = 1 \cdots L$

Base step:

When $i = L-1$,

$$b_{L-1}(s) = P[x_L \mid s_{L-1} = s]$$



$$= P[s' \mid s] * P[x_L \mid s'] + P[s' \mid s] * P[x_L \mid s'] + P[s' \mid s] * P[x_L \mid s'] + \cdots$$

$$= \sum_{s'} P[s' \mid s] * P[x_L \mid s'] = \sum_{s'} \left(P[s' \mid s] * P[x_L \mid s] * b_L(s') \right)$$

($b_L(s') = 1$)

$P[s' \mid s]$: Transition probability

$P[x_L \mid s']$: Emission Probability or observation prob.

$$b_i(s) = \sum_{s'} \left(P[s' \mid s] * P[x_{i+1} \mid s'] * b_{i+1}(s') \right)$$

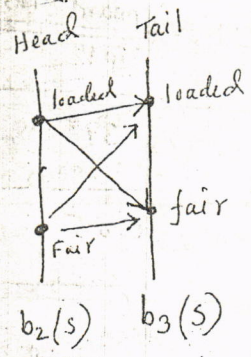
Transition prob

Observation probability

backward probability

Find $P[s_i = \text{loaded} \mid \text{Head Head Tail}]$ and $P[s_i = \text{fair} \mid \text{Head Head Tail}]$ using backward algorithm.

Ans:



Base step

$$b_3(\text{loaded}) = 1$$

$$b_3(\text{fair}) = 1$$

Backward step-1

$$b_2(\text{loaded}) = P(\text{loaded} \mid \text{loaded}) * P(\text{Tail} \mid \text{loaded}) * b_3(\text{loaded}) + P(\text{fair} \mid \text{loaded}) * P(\text{Tail} \mid \text{fair}) * b_3(\text{fair})$$

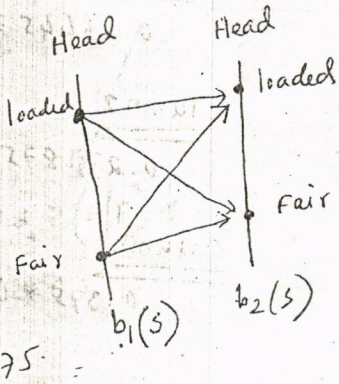
$$= 0.9 * 0.25 * 1 + 0.1 * 0.5 * 1$$

$$= 0.275$$

$$b_2(\text{fair}) = P(\text{loaded} \mid \text{Fair}) * P(\text{Tail} \mid \text{loaded}) * b_3(\text{loaded}) + P(\text{fair} \mid \text{fair}) * P(\text{Tail} \mid \text{fair}) * b_3(\text{fair})$$

$$= 0.1 * 0.25 * 1 + 0.9 * 0.5 * 1$$

$$= 0.475$$



Backward step-2

$$b_1(\text{loaded}) = P(\text{loaded} \mid \text{loaded}) * P(\text{Head} \mid \text{loaded}) * b_2(\text{loaded}) + P(\text{Fair} \mid \text{loaded}) * P(\text{Head} \mid \text{Fair}) * b_2(\text{fair})$$

$$= 0.9 * 0.75 * 0.275 + 0.1 * 0.5 * 0.475$$

$$= 0.209$$

$$b_1(\text{fair}) = P(\text{loaded} \mid \text{Fair}) * P(\text{Head} \mid \text{loaded}) * b_2(\text{loaded}) + P(\text{fair} \mid \text{Fair}) * P(\text{Head} \mid \text{Fair}) * b_2(\text{Fair})$$

$$= 0.1 * 0.75 * 0.275 + 0.9 * 0.5 * 0.475 = 0.234$$

Ques: Find likelihood of {Head, Head, Tail}

Ans: From Forward Pass:

	step1	step2	step3
loaded	0.375	0.271875	0.06445
Fair	0.25	0.13125	0.07265

From Backward Pass:

	step2	step1	base step
loaded	0.209	0.275	1
Fair	0.234	0.475	1

Likelihood

Column 3

$0.06445 * 1 + 0.07265 * 1 = 0.1371$

Column 2

$0.271875 * 0.275 + 0.13125 * 0.475 = 0.1371$

Column 1

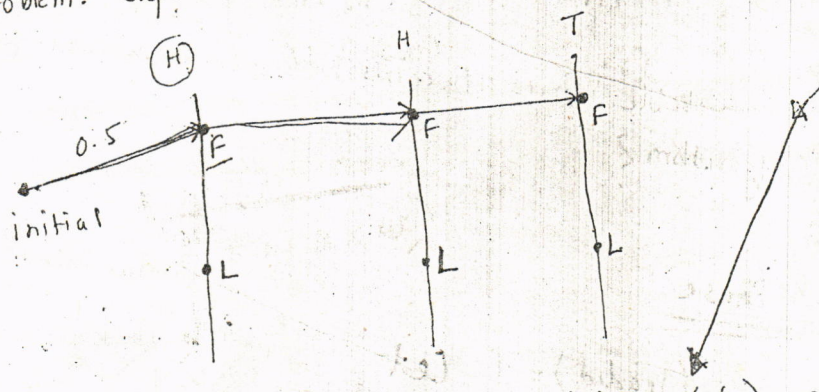
$0.375 * 0.209 + 0.25 * 0.234 = 0.1371$

What is the most probable series of coins to observe sequence {Head, Head, Tail}

Ans:

S_1 S_2 S_3	$P[\text{Head Head Tail}, S_1, S_2, S_3]$
F F F	$(0.5)^3 * 0.5 * (0.9)^2 = 0.050625$
F F L	$(0.5)^2 * 0.25 * 0.5 * 0.9 * 0.1 = 0.0028125$
F L F	$0.5 * 0.75 * 0.5 * 0.5 * 0.1 * 0.1 = 0.0009375$
F L L	$0.5 * 0.75 * 0.25 * 0.5 * 0.1 * 0.9 = 0.00422$
L F F	$0.75 * 0.5 * 0.5 * 0.5 * 0.1 * 0.9 = 0.0084375$
L F L	$0.75 * 0.5 * 0.25 * 0.5 * 0.1 * 0.1 = 0.000468$
L L F	$0.75 * 0.75 * 0.5 * 0.5 * 0.9 * 0.1 = 0.01265$
L L L	$0.75 * 0.75 * 0.25 * 0.5 * 0.9 * 0.9 = 0.0569$ → max

Problem: exponential in length of observation



$0.5 * P(H|F) * P(F|F) * P(H|F) * P(F|F) * P(T|F)$

$= 0.5 * 0.5 * 0.9 * 0.5 * 0.9 * 0.5$

$= (0.5)^3 * 0.5 * (0.9)^2$

Viterbi algorithm: (Backward Starting - Last 2 Start)

Backward phase:

calculate $v_i(s) = P^*[X_{i+1} \dots X_L \mid S_i = s]$

Base:

$$V_L(s) = 1$$

step i:

$$v_i(s) = \max_{s'} \left\{ P(s'|s) * P(X_{i+1}|s') * v_{i+1}(s') \right\}$$

(Transition probability, observation probability, backward probability)

$$\pi_{i+1}(s) = \text{ARGMAX}_{s'} \left\{ P(s'|s) * P(X_{i+1}|s') * v_{i+1}(s') \right\}$$

Forward phase:

Trace path of maximum probability

Base: $\pi_1 = s_1^* = \text{ARGMAX}_{s'} \left\{ P(s') * P(X_1|s') \, v_1(s') \right\}$

step: $s_{i+1}^* = \pi_{i+1}(s_i^*)$

What is the most probable series of coins to observe sequence {Head, Head, Tail}

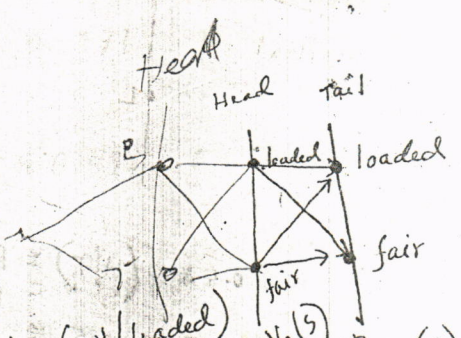
using Viterbi algorithm?

Ans:

Backward Phase

base: $v_3(\text{loaded}) = 1$ let

$v_3(\text{fair}) = 1$



States

0

=3

=2

=1

sequence

step1:

$$v_2(\text{loaded}) = \max \left(P(\text{loaded}|\text{loaded}) * P(\text{Tail}|\text{loaded}) * v_3(\text{loaded}), \; P(\text{fair}|\text{loaded}) * P(\text{Tail}|\text{fair}) * v_3(\text{fair}) \right)$$

$$= \max(0.9 * 0.25 * 1, \; 0.1 * 0.5 * 1)$$

$$= 0.225$$

$\pi_3(\text{loaded}) = \text{loaded}$

$\pi_3(\text{loaded}) =$ Which one is max

$$V_2(\text{Fair}) = \max\Big(P[\text{loaded}|\text{fair}] * P[\text{Tail}|\text{loaded}] * V_3(\text{loaded}),$$
$$P[\text{fair}|\text{fair}] * P[\text{Tail}|\text{fair}] * V_3(\text{fair}) \Big)$$

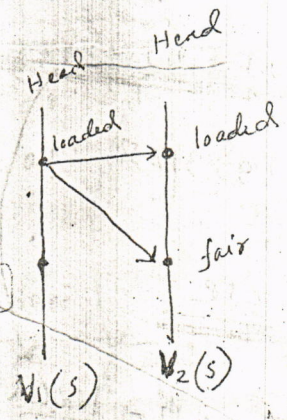
$$= \max(0.1 * 0.25 * 1,\ 0.9 * 0.5 * 1)$$

$$= 0.45$$

$$\pi_3(\text{Fair}) = \text{Fair}$$

Step 2 :

$$V_1(\text{loaded}) = \max\Big(P[\text{loaded}|\text{loaded}] * P[\text{Head}|\text{loaded}] * V_2(\text{loaded}),$$
$$P[\text{fair}|\text{loaded}] * P[\text{Head}|\text{fair}] * V_2(\text{fair}) \Big)$$

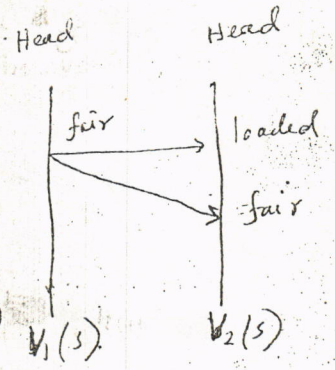


$$= \max(0.9 * 0.75 * 0.225,\ 0.1 * 0.5 * 0.45)$$

$$= 0.151875$$

$$\pi_2(\text{loaded}) = \text{loaded}$$

$$V_1(\text{fair}) = \max\Big(P[\text{loaded}|\text{fair}] * P(\text{Head}|\text{loaded}) * V_2(\text{loaded}),$$
$$P[\text{fair}|\text{fair}] * P(\text{Head}|\text{fair}) * V_2(\text{fair}) \Big)$$

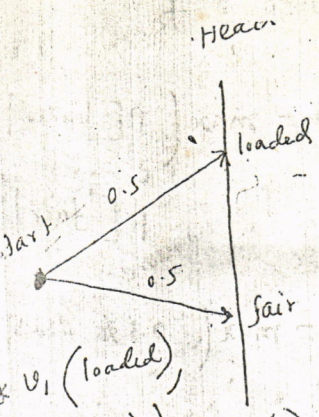


$$= \max(0.1 * 0.75 * 0.225,\ 0.9 * 0.5 * 0.45)$$

$$= 0.2025$$

$$\pi_2(\text{Fair}) = \text{Fair}$$

step3:



P(Head Head Tail)

= ~~max(P(loaded) * P(loaded)~~

= max(P(loaded) * P(Head | loaded) * V_1(loaded),
P(fair) * P(Head | fair) * V_1(fair))

= max(0.5 * 0.75 * 0.1875, 0.5 * 0.5 * 0.2025)

= 0.0703125

Forward phase

$\pi_1^* = s_1^*$ = loaded

$s_2^* = \pi_2(s_1^*)$

= π_2(loaded)

= loaded

$s_3^* = \pi_3(s_2^*)$

= π_3(loaded)

= loaded

Most Probable coin sequence = $\{s_1^* s_2^* s_3^*\}$

= {loaded loaded loaded}

π_3(Head) = L

π_3(F) = F

π_2(L) = L

π_2(F) = F

π_1(L) = L

$\pi_1 = s_1^*$ = F

$s_2^* = \pi_2$(F)

= L

$s_3^* = \pi_3$(L)

= F

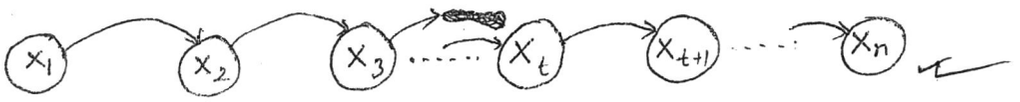
F L F L F

Dr. Md. Nurul Huda
Markov Model
MSCSE, Advanced AI

Markov Process:

Markov Property:

The state of the system at time t+1 depends only on the state of the system at time t.



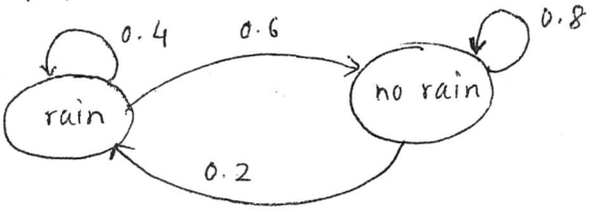
Example1

Weather:

Raining today → 40% rain tomorrow
Raining today → 60% no rain tomorrow

Not raining today → 20% rain tomorrow
Not raining today → 80% no rain tomorrow

Stochastic Finite State Machine (FSM):



The transition matrix:

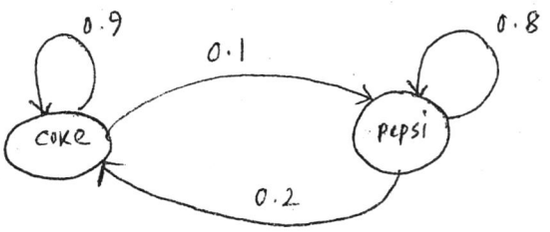
$$P = \begin{array}{c} \\ \text{Rain} \\ \text{No Rain} \end{array} \begin{array}{c} \begin{array}{cc} \text{Rain} & \text{No Rain} \end{array} \\ \begin{bmatrix} 0.4 & 0.6 \\ 0.2 & 0.8 \end{bmatrix} \end{array}$$

↑ stochastic matrix since rows sum up to 1.

[Rows and columns sum up to 1]

Example 2:

- Given that a person's last cola purchase was coke, there is a 90% chance that his next cola purchase will also be coke.
- If a person's last cola purchase was pepsi, there is an 80% chance that his next ~~data~~ cola purchase will also be pepsi.



Transition matrix:

$$P = \begin{array}{c} \\ \text{coke} \\ \text{pepsi} \end{array} \begin{array}{c} \begin{array}{cc} \text{coke} & \text{pepsi} \end{array} \\ \begin{bmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{bmatrix} \end{array}$$

Problem1:

Given that a person is currently a pepsi purchaser, what is the probability that he will purchase coke two purchases from now

Ans:

Method1: $P[\text{Pepsi} \rightarrow ? \rightarrow \text{Coke}] = P[\text{Pepsi} \rightarrow \text{Coke} \rightarrow \text{Coke}] + P[\text{Pepsi} \rightarrow \text{Pepsi} \rightarrow \text{Coke}]$

$$= P[\text{coke}|\text{coke}] * P[\text{coke}|\text{Pepsi}] + P[\text{coke}|\text{Pepsi}] * P[\text{Pepsi}|\text{Pepsi}]$$

$$= 0.9 * 0.2 + 0.2 * 0.8$$

$$= 0.34 \quad \text{Ans:}$$

Method2:

$$P = \begin{bmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{bmatrix}$$

$$P = P \times P = \begin{bmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{bmatrix} \times \begin{bmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{bmatrix}$$

$$= \begin{array}{c} \\ \text{coke} \\ \text{pepsi} \end{array} \begin{array}{c} \begin{array}{cc} \text{coke} & \text{pepsi} \end{array} \\ \begin{bmatrix} 0.83 & 0.17 \\ 0.34 & 0.66 \end{bmatrix} \end{array}$$

$$P[\text{Pepsi} \rightarrow 2 \rightarrow \text{coke}] = 0.34$$

Problem 2 :

Given that a person is currently a coke purchaser, what is the probability that he will purchase pepsi three purchases from now ?

Method 1 :

$$P[\text{coke} \to ? \to ? \to \text{pepsi}] = P[\text{coke} \to \text{Pepsi} \to \text{Pepsi} \to \text{Pepsi}] + P[\text{coke} \to \text{Pepsi} \to \text{Coke} \to \text{Pepsi}] + P[\text{coke} \to \text{Coke} \to \text{Pepsi} \to \text{Pepsi}] + P[\text{coke} \to \text{Coke} \to \text{Coke} \to \text{Pepsi}]$$

$$= P[\text{pepsi}|\text{pepsi}] * P[\text{pepsi}|\text{pepsi}] * P[\text{pepsi}|\text{coke}] + P[\text{pepsi}|\text{coke}] * P[\text{coke}|\text{pepsi}] * P[\text{pepsi}|\text{coke}] + P[\text{pepsi}|\text{pepsi}] * P[\text{pepsi}|\text{coke}] * P[\text{coke}|\text{coke}] + P[\text{pepsi}|\text{coke}] * P[\text{coke}|\text{coke}] * P[\text{coke}|\text{coke}]$$

$$= 0.8 * 0.8 * 0.1 + 0.1 * 0.2 * 0.1 + 0.8 * 0.1 * 0.9 + 0.1 * 0.9 * 0.9$$

$$= 0.064 + 0.002 + 0.072 + 0.081$$

$$= 0.219.$$

Method 2 :

$$P^3 = P \times P \times P$$

$$= \begin{bmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{bmatrix} \times \begin{bmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{bmatrix} \times \begin{bmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{bmatrix}$$

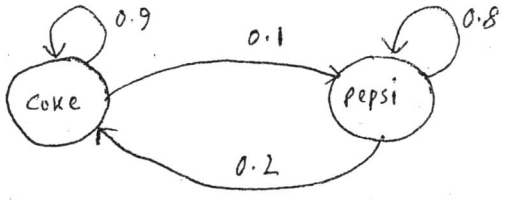
$$= \begin{bmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{bmatrix} \times \begin{bmatrix} 0.83 & 0.17 \\ 0.34 & 0.66 \end{bmatrix}$$

$$= \begin{matrix} \\ \text{coke} \\ \text{pepsi} \end{matrix} \begin{matrix} \begin{matrix} \text{coke} & \text{Pepsi} \end{matrix} \\ \begin{bmatrix} 0.781 & 0.219 \\ 0.438 & 0.562 \end{bmatrix} \end{matrix}$$

$$P[\text{coke} \to ? \to ? \to \text{pepsi}] = 0.219$$

Problem 3

Assume each person makes one cola purchase per week. Suppose 60% of all people now drink coke, and 40% drink pepsi. What fraction of people will be drinking coke three weeks from now?



Method 1:

$$P^3 = P \times P \times P$$

$$= \begin{bmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{bmatrix} \times \begin{bmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{bmatrix} \times \begin{bmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{bmatrix}$$

$$= \begin{bmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{bmatrix} \times \begin{bmatrix} 0.83 & 0.17 \\ 0.34 & 0.66 \end{bmatrix}$$

$$= \begin{matrix} \\ \text{Coke} \\ \text{Pepsi} \end{matrix} \begin{matrix} \begin{matrix} \text{Coke} & \text{Pepsi} \end{matrix} \\ \begin{bmatrix} 0.781 & 0.219 \\ 0.438 & 0.562 \end{bmatrix} \end{matrix}$$

$$P[\text{Coke} \to ? \to ? \to \text{Coke}] = 0.781$$

$$P[\text{Pepsi} \to ? \to ? \to \text{Coke}] = 0.438$$

Fraction of people = 60% of $P[\text{Coke} \to ? \to ? \to \text{Coke}]$

$+$ 40% of $P[\text{Pepsi} \to ? \to ? \to \text{Coke}]$

$= 0.6 * 0.781 + 0.4 * 0.438$

$= 0.6438$

Markov Models

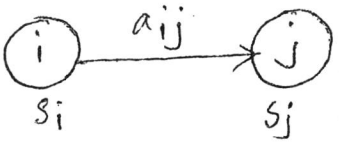
- set of states: $\{s_1, s_2, \cdots s_k, s_N\}$
- Markov chain property: probability of each subsequent state depends only on what was the previous state.

$$P(s_k \mid s_1 \cdots s_{k-1}) = P(s_k \mid s_{k-1})$$

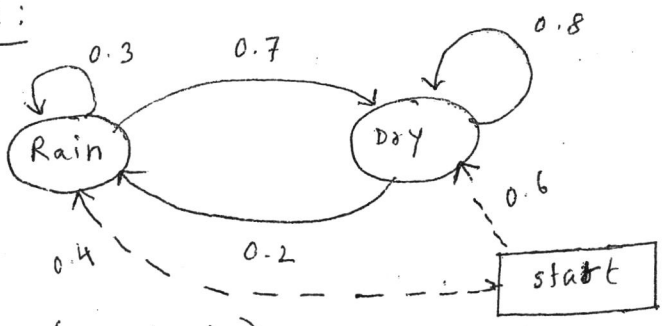
- Markov model:

Transition probabilities, $a_{ij} = P(s_j \mid s_i)$

Initial probabilities, $\pi_i = P(s_i)$



Example:



$P(\text{Rain} \mid \text{Rain}) = 0.3$

$P(\text{Dry} \mid \text{Rain}) = 0.7$

$P(\text{Rain} \mid \text{Dry}) = 0.2$

$P(\text{Dry} \mid \text{Dry}) = 0.8$

Find $P(\text{Dry}, \text{Dry}, \text{Rain}, \text{Rain}) = P(\text{Rain} \mid \text{Rain}) * P(\text{Rain} \mid \text{Dry}) * P(\text{Dry} \mid \text{Dry}) * P(\text{Dry})$

$= 0.3 * 0.2 * 0.8 * 0.6$

$= 0.06 * 0.48$

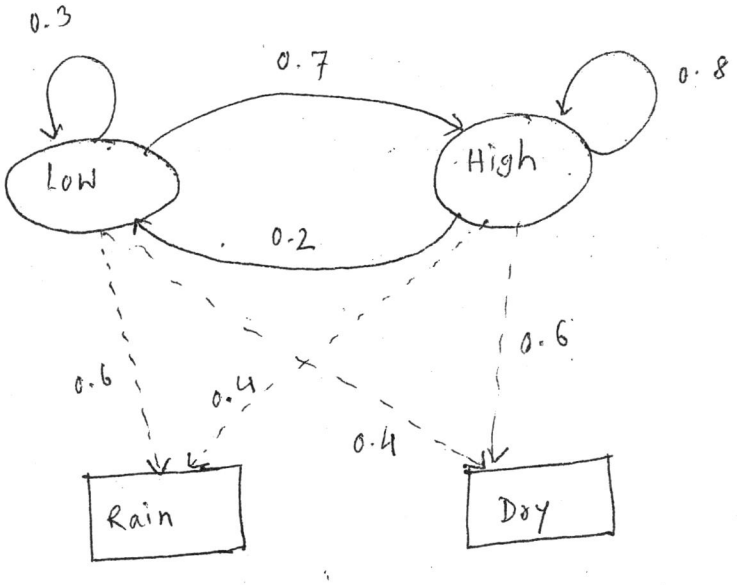
$= 0.0288$

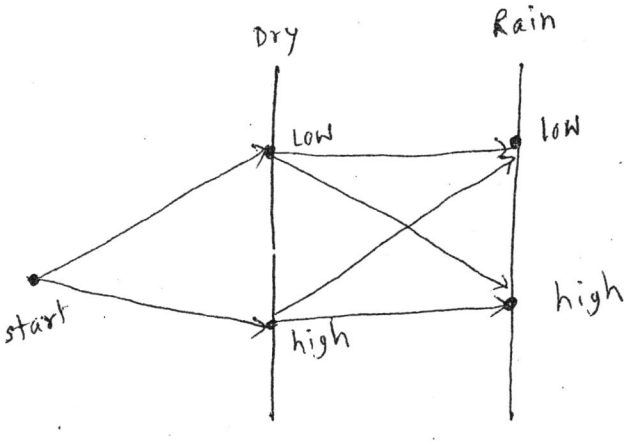
~~$P(S_k | S_1 S_2 \cdots S_{k-1})$ value:~~

$$P(S_1, S_2, S_3, \ldots, S_k) = P(S_k | S_1 S_2 \cdots S_{k-1}) P(S_1 S_2 \cdots S_{k-1})$$

$$= P(S_k | S_{k-1}) * P(S_{k-1} | S_{k-2}) P(S_1 S_2 \cdots S_{k-2})$$

$$= P(S_k | S_{k-1}) P(S_{k-1} | S_{k-2}) \cdots P(S_2 | S_1) P(S_1)$$





$$P(\text{Dry, Rain}) = P(\text{low}) * P(\text{Dry} | \text{low}) * P(\text{low} | \text{low}) * P(\text{Rain} | \text{low})$$
$$+ P(\text{low}) * P(\text{Dry} | \text{low}) * P(\text{high} | \text{low}) * P(\text{Rain} | \text{high})$$
$$+ P(\text{high}) * P(\text{Dry} | \text{high}) * P(\text{low} | \text{high}) * P(\text{Rain} | \text{low})$$
$$+ P(\text{high}) * P(\text{Dry} | \text{high}) * P(\text{high} | \text{high}) * P(\text{Rain} | \text{high})$$

$$= 2$$